STA 4210 HW 3

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1.(R) In this problem, you will be working with the `cars` dataset, which contains two variables: - `speed`: The speed of the car (in mph) - `dist`: The stopping distance (in feet)

```
library(car)
car.reg <- lm(dist ~ speed, data = cars)

e <- rstudent(car.reg)
shapiro.test(e)  # Shapiro-Wilk
```

```
##
##  Shapiro-Wilk normality test
##
## data:  e
## W = 0.93418, p-value = 0.00798
```

```
ncvTest(car.reg)  # Breusch-Pagan
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 4.650233, Df = 1, p = 0.031049
```

a. (5pt) Based on the results of the significance test, there might be violation to the Homogeneity assumption and Normal assumption. Use the `boxcox` function from the `MASS` package to find the optimal value of the Box-Cox transformation parameter (λ).

Solutions:

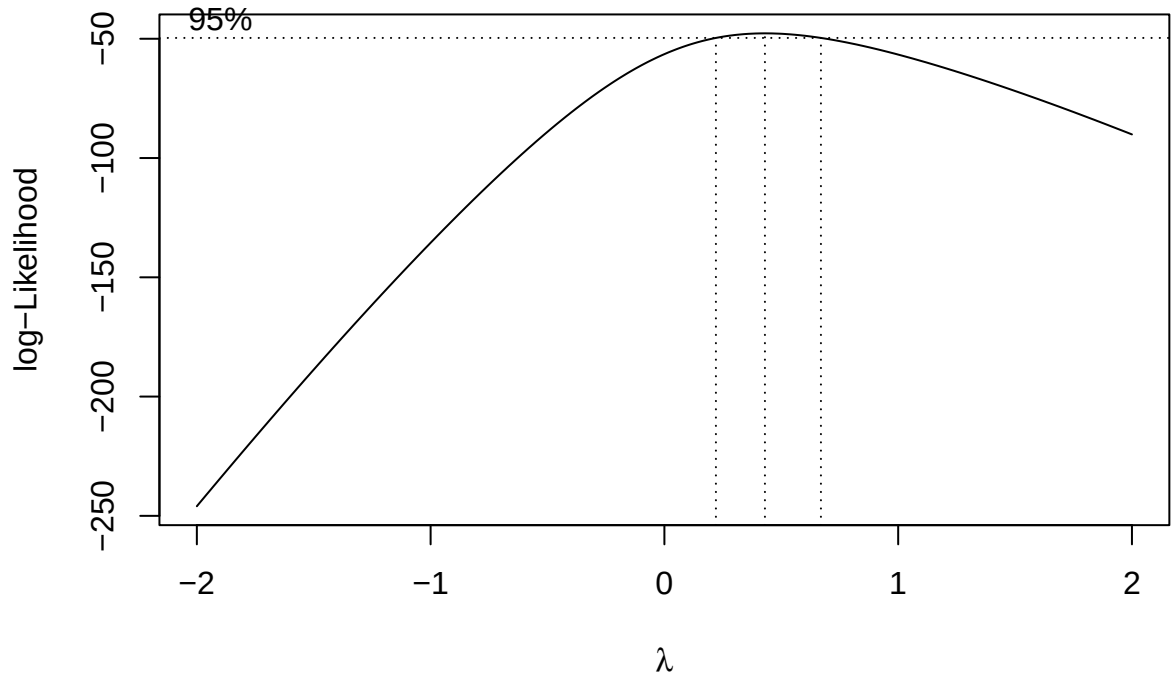
The Box–Cox transformation for a positive response variable Y is defined as

$$Y^{(\lambda)} = \begin{cases} \dfrac{Y^{\lambda} - 1}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases}$$

The optimal value of λ is chosen to maximize the log-likelihood function.

```
library(MASS)

# Box-Cox on the ORIGINAL model
bc <- boxcox(car.reg, lambda = seq(-2, 2, by = 0.01), plotit = TRUE)
```




```r
# Optimal lambda = argmax of log-likelihood
lambda_opt <- bc$x[which.max(bc$y)]
lambda_opt
```

```
## [1] 0.43
```

The Box–Cox analysis suggests an optimal transformation parameter of $\hat{\lambda} = 0.43$. Since this value is close to 0.5, the transformation is approximately a square-root transformation, which may help stabilize the variance and improve the normality of the residuals.

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b. (5pt) Apply the transformation to the response variable using the optimal λ and refit the model using the transformed data.

Solution:

The transformed response variable is computed using the Box–Cox formula

$$Y^{(\lambda)} = \frac{Y^{\lambda} - 1}{\lambda}, \quad \lambda \neq 0.$$

Substituting $\hat{\lambda} = 0.43$,

$$Y^{(0.43)} = \frac{Y^{0.43} - 1}{0.43}.$$

```r
lambda_opt <- 0.43

# Transform response
dist_bc <- (cars$dist^lambda_opt - 1)/lambda_opt

# Refit regression model
car.reg.bc <- lm(dist_bc ~ speed, data = cars)

summary(car.reg.bc)
```

```
##
## Call:
## lm(formula = dist_bc ~ speed, data = cars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1616 -1.0665 -0.3097  0.8215  4.8550
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.05004    0.75373   1.393     0.17
## speed        0.50538    0.04634  10.906 1.37e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.715 on 48 degrees of freedom
## Multiple R-squared:  0.7125, Adjusted R-squared:  0.7065
## F-statistic: 118.9 on 1 and 48 DF,  p-value: 1.373e-14
```

From the output: The intercept = 1.05004 slope = 0.50538 Thus:

$$\widehat{Y^{(0.43)}} = 1.05004 + 0.50538(x).$$

The response variable was transformed using the optimal Box–Cox parameter $\hat{\lambda} = 0.43$, and the regression model was refit using the transformed response. The fitted model shows a significant positive relationship between speed and the transformed stopping distance ($p < 0.001$).

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c. (10pt) Check ALL model assumptions by significance tests with $\alpha = 0.05$ after applying the transformation. Are there any potential violations?

Solution:

After applying the Box–Cox transformation with $\hat{\lambda} = 0.43$, the regression assumptions were evaluated using significance tests at $\alpha = 0.05$.

The following tests were performed based on the previous HW:

1.Shapiro–Wilk test (normality)

2.Runs test (randomness)

3.Durbin–Watson test (independence)

4.Levene's test (homogeneity of variance)

5.Breusch–Pagan / Cook–Weisberg test (heteroscedasticity)

6.Lack-of-fit test (model adequacy / linearity)

```r
if (!require(randtests)) install.packages("randtests")
if (!require(lmtest)) install.packages("lmtest")
if (!require(car)) install.packages("car")

library(randtests)
library(lmtest)
library(car)


# Studentized residuals
e_bc <- rstudent(car.reg.bc)

### 1. Shapiro-Wilk test (Normality)
shapiro.test(e_bc)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  e_bc
## W = 0.97421, p-value = 0.3403
```

```r
### 2. Runs test (Independence/if the residual is random)
runs.test(e_bc)
```

```
##
##  Runs Test
##
## data:  e_bc
## statistic = 0.57155, runs = 28, n1 = 25, n2 = 25, n = 50, p-value =
## 0.5676
## alternative hypothesis: nonrandomness
```

```r
### 3. Durbin-Watson test (Independence)
durbinWatsonTest(car.reg.bc)
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1     0.001743558      1.959577   0.746
##  Alternative hypothesis: rho != 0
```

```r
### 4. Levene's Test (Homogeneity)
leveneTest(e_bc ~ as.factor(cars$speed))
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group 18  1.2497 0.2849
##       31
```

```r
### 5. Breusch-Pagan / Cook-Weisberg (Homogenetiy)
bptest(car.reg.bc)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  car.reg.bc
## BP = 0.11168, df = 1, p-value = 0.7382
```

```r
### 6. Lack-of-Fit Test (linearity) Compare linear vs
### saturated model
car.reg.full <- lm(dist_bc ~ factor(speed), data = cars)
anova(car.reg.bc, car.reg.full)
```

```
## Analysis of Variance Table
##
## Model 1: dist_bc ~ speed
## Model 2: dist_bc ~ factor(speed)
##   Res.Df     RSS Df Sum of Sq      F Pr(>F)
## 1     48 141.211
## 2     31  92.394 17    48.817 0.9635 0.5177
```

Shapiro–Wilk test (Normality)

$$H_0 : \text{Residuals are normally distributed}$$

The p-value is $0.3403 > 0.05$, so we fail to reject H_0. There is no evidence against the normality assumption.

Runs test (Independence/Randomness)

$$H_0 : \text{Residuals occur in random order}$$

The p-value is $0.5676 > 0.05$, so we fail to reject H_0. The residuals appear to occur randomly and are independent.

Durbin–Watson test (Independence)

$$H_0 : \text{Errors are independent}$$

The p-value is $0.796 > 0.05$, so we fail to reject H_0. There is no evidence of autocorrelation among the residuals.

Levene's test (Homogeneity of variance)

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

The p-value is $0.2849 > 0.05$, so we fail to reject H_0. The variances appear to be homogeneous.

Breusch–Pagan test (Constant variance)

$$H_0 : \sigma^2 \text{ is constant}$$

The p-value is $0.7382 > 0.05$, so we fail to reject H_0. There is no evidence of heteroscedasticity.

Lack-of-fit test (Model adequacy)

$$H_0 : \text{The linear model is adequate (no lack of fit)}$$

The p-value is $0.5177 > 0.05$, so we fail to reject H_0. The linear model appears to provide an adequate fit to the data.

Overall Conclusion

Since all p-values are greater than $\alpha = 0.05$, there is no evidence of violations of the regression assumptions after applying the Box–Cox transformation. Therefore, the transformed model satisfies the assumptions of normality, independence, homogeneity of variance, randomness, and linear model adequacy.

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d. (5pt) For the refitted model with the transformed response, compute simultaneous confidence intervals for the mean transformed stopping distance at speed = 20 mph and 24 mph using the Bonferroni method with at least 95% coverage. You can use the `predict` function.

Solution:

```r
# Bonferroni simultaneous confidence intervals for the MEAN
# transformed response at speed = 20 and 24 mph (overall
# coverage >= 95%)

newdata <- data.frame(speed = c(20, 24))
g <- nrow(newdata)  # number of intervals
alpha <- 0.05
level_bonf <- 1 - alpha/g  # Bonferroni per-interval confidence level = 0.975

pred_bonf <- predict(car.reg.bc, newdata = newdata, interval = "confidence",
    level = level_bonf)

# Display results with speeds
cbind(newdata, pred_bonf)
```

```
##   speed      fit      lwr      upr
## 1    20 11.15758 10.41038 11.90479
## 2    24 13.17909 12.09957 14.25861
```

We construct simultaneous confidence intervals for the mean transformed stopping distance at speeds 20 mph and 24 mph using the Bonferroni method with overall confidence level at least 95%.

Since two intervals are required ($g = 2$) and $\alpha = 0.05$, the Bonferroni per-interval confidence level is

$$1 - \frac{\alpha}{g} = 1 - \frac{0.05}{2} = 0.975.$$

Thus, each individual confidence interval is constructed at the 97.5% confidence level to guarantee at least 95% simultaneous coverage.

Using the fitted transformed model and the `predict` function, the simultaneous confidence intervals for the mean transformed stopping distance are:

For speed = 20 mph:

$$(10.4104,\ 11.9048)$$

For speed = 24 mph:

$$(12.0996,\ 14.2586)$$

These intervals provide at least 95% simultaneous confidence that the true mean transformed stopping distances at speeds 20 mph and 24 mph lie within their respective intervals.

2.(R) In this problem, you will work with matrices in R. You can use the following R functions to complete the tasks: - `matrix()` to generate a matrix - `%*%` for matrix multiplication - `det()` to calculate the determinant - `qr()` to find the rank of a matrix - `t()` for the transpose of a matrix

a. (5pt) Let A and B be the following matrices:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 5 & 6 & -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 4 \end{pmatrix}$$

Multiply matrix A by matrix B (i.e., compute AB).

Solution:

```r
# Define matrices A and B

A <- matrix(c(2, -1, 5, 1, 3, 6, 0, 4, -2), nrow = 3, byrow = FALSE)

B <- matrix(c(1, -1, 3, 2, 0, 4), nrow = 3, byrow = FALSE)

# Matrix multiplication AB
AB <- A %*% B

AB
```

```
##      [,1] [,2]
## [1,]    1    4
## [2,]    8   14
## [3,]   -7    2
```

We are given the matrices

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 5 & 6 & -2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 4 \end{pmatrix}.$$

The matrix product AB is computed by multiplying rows of A with columns of B.

$$AB = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 5 & 6 & -2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 8 & 14 \\ -7 & 2 \end{pmatrix}.$$

Thus, the matrix product is

$$AB = \begin{pmatrix} 1 & 4 \\ 8 & 14 \\ -7 & 2 \end{pmatrix}.$$

---

b. (5pt) Calculate the transpose of matrix A (denoted A^T). Is matrix $A^T A$ symmetric? Verify this by checking whether it is equal to its transpose. ## solution:

```r
# Matrix A from part (a)
A <- matrix(c(2, -1, 5, 1, 3, 6, 0, 4, -2), nrow = 3, byrow = FALSE)

# Transpose of A
A_T <- t(A)
A_T
```

```
##      [,1] [,2] [,3]
## [1,]    2   -1    5
## [2,]    1    3    6
## [3,]    0    4   -2
```

```r
# Compute A^T A
ATA <- A_T %*% A
ATA
```

```
##      [,1] [,2] [,3]
## [1,]   30   29  -14
## [2,]   29   46    0
## [3,]  -14    0   20
```

```r
# Check if symmetric: compare with its transpose
is_symmetric <- all(ATA == t(ATA))
is_symmetric
```

```
## [1] TRUE
```

The transpose of matrix A is obtained by interchanging rows and columns.

Given

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 5 & 6 & -2 \end{pmatrix},$$

the transpose is

$$A^T = \begin{pmatrix} 2 & -1 & 5 \\ 1 & 3 & 6 \\ 0 & 4 & -2 \end{pmatrix}.$$

Next, we compute $A^T A$:

$$A^T A = \begin{pmatrix} 30 & 29 & -14 \\ 29 & 46 & 0 \\ -14 & 0 & 20 \end{pmatrix}.$$

A matrix is symmetric if it equals its transpose. Since

$$(A^T A)^T = A^T A,$$

the matrix $A^T A$ is symmetric. This was also verified in R by checking `all(ATA == t(ATA))`, which returned TRUE.

---

c. (5pt) Consider the matrix C:

$$C = \begin{pmatrix} 3 & 1 \\ 5 & 7 \end{pmatrix}$$

Calculate the determinant of matrix C. What does the determinant tell you about the matrix?

```r
# Define matrix C
C <- matrix(c(3, 5, 1, 7), nrow = 2, byrow = FALSE)

# Determinant of C
det_C <- det(C)
det_C
```

```
## [1] 16
```

We are given the matrix

$$C = \begin{pmatrix} 3 & 1 \\ 5 & 7 \end{pmatrix}.$$

The determinant of a 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is computed as

$$\det(C) = ad - bc.$$

Thus,

$$\det(C) = (3)(7) - (1)(5) = 21 - 5 = 16.$$

Since the determinant is nonzero ($\det(C) = 16 \neq 0$), the matrix is invertible and has full rank. This means its columns (and rows) are linearly independent, and the matrix has an inverse.

---

d. (5pt) Using matrix A from part (a). Calculate the rank of matrix A. Is matrix A full-rank (i.e., does it have the maximum rank)? *R Hint*: You can use `qr()` function obtain the rank directly, or use `eigen()` function to find the eigenvalues and rank is the number of non-zero eigenvalues.

```r
# Matrix A from part (a)
A <- matrix(c(2, -1, 5, 1, 3, 6, 0, 4, -2), nrow = 3, byrow = FALSE)

# Rank using QR decomposition
rank_A <- qr(A)$rank
rank_A
```

```
## [1] 3
```

```r
# Check full-rank (maximum rank for 3x3 is 3)
is_full_rank <- (rank_A == min(dim(A)))
is_full_rank
```

```
## [1] TRUE
```

Using matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 5 & 6 & -2 \end{pmatrix},$$

we compute the rank of A in R using the QR decomposition via `qr(A)$rank`.

The result is

$$\text{rank}(A) = 3.$$

Since A is a 3×3 matrix, its maximum possible rank is

$$\min(3, 3) = 3.$$

Because $\text{rank}(A) = 3$, matrix A is full-rank. Equivalently, its columns are linearly independent, and A is invertible.

---

e. (5pt) If a square matrix is full-rank, its inverse can be computed. Consider the following matrix D:

$$D = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$$

Check if matrix D is full-rank. If D is full-rank, calculate its inverse. *R Hint*: Use `solve()` to calculate the inverse of a matrix. Make sure to check the rank first.

```r
# Define matrix D
D <- matrix(c(4, 2, 7, 6), nrow = 2, byrow = FALSE)

# Rank of D
rank_D <- qr(D)$rank
rank_D
```

```
## [1] 2
```

```r
# Check if full-rank
is_full_rank <- (rank_D == min(dim(D)))
is_full_rank
```

```
## [1] TRUE
```

```r
# Inverse of D (only if full-rank)
D_inv <- solve(D)
D_inv
```

```
##      [,1] [,2]
## [1,]  0.6 -0.7
## [2,] -0.2  0.4
```

We are given

$$D = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}.$$

First, we check whether D is full-rank. The rank of a 2×2 matrix can be at most 2. Using R, we obtain

$$\text{rank}(D) = 2.$$

Since the rank equals the maximum possible rank, matrix D is full-rank and therefore invertible.

The inverse of a 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is given by

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant of D is

$$\det(D) = (4)(6) - (7)(2) = 24 - 14 = 10.$$

Thus,

$$D^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}.$$

Therefore, matrix D is full-rank and its inverse exists.